Machine Learning Surrogates of Large-Eddy Simulation for Pedestrian-Level Wind in Urban Canopies

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***Abstract*—Large-eddy simulation (LES) resolves pedestrian-level wind in urban canopies but is too expensive for design studies that compare many layouts. We assess how well classical machine learning approximates it on all 538 UrbanTALES cases of idealized building arrays and real neighbourhoods in 22 countries. An audit of the released files found that the building rasters are vertically flipped relative to the wind fields (open-ground agreement 0.71 as stored, 0.99 after a flip), that some files store values inside buildings and some open-ground cells hold a speed of exactly zero (together biasing the mean wind of 59 development cases by up to 14.6%), and that the friction velocity, which sets the wind scale, differs between cases; a single loader corrects all of them. On held-out city and layout groups, under nested grouped cross-validation, the most accurate model, LightGBM, predicts the normalised open-ground mean wind with $R^2 = 0.876$ (idealized 0.961, realistic 0.787) and exceeds a nine-feature random-forest baseline by 0.135 in R^2; a tuned random forest also exceeds it, while the gains of XGBoost and a Gaussian process are not significant. On a frozen test set of 53 cases, the non-linear models reach R^2 of 0.857 to 0.931 against 0.827 for the baseline. A pixel-level tree model recovers the level of each wind field but only part of its spatial pattern (median per-case R^2 0.418 idealized, 0.247 realistic) and underpredicts high-percentile speeds. A random case split changes the score on held-out realistic cases little, and a far less regularised model does not improve it, which indicates that the hand-crafted local features limit tabular field prediction and motivates convolutional models in Phase 2.**

***Index Terms*—Gradient boosting, grouped cross-validation, large-eddy simulation, pedestrian-level wind, Shapley additive explanations, surrogate modelling, urban canopy**

I. Introduction

PEDESTRIAN-LEVEL wind governs outdoor comfort, the removal of pollutants and heat from street canyons, and wind safety around tall buildings [1]. Computational fluid dynamics (CFD) resolves these effects, and large-eddy simulation (LES) resolves the turbulent eddies that carry them, but at metre resolution it is computationally expensive and requires careful set-up and validation [2]. Design studies that compare many layouts therefore cannot afford one simulation per option. A surrogate model trained on completed simulations can replace the simulation in such studies if it generalises to layouts it has not seen.

UrbanTALES [3] released 538 LES cases of pedestrian-level wind over idealized arrays and real neighbourhoods. The first surrogate trained on these data, a U-Net [4], was evaluated on a random split of cases, so neighbourhoods from the same city can appear in both the training and the test sets. Apart from a random-forest baseline on a few layout descriptors in [4], classical learners on physically motivated morphometric features have not, to our knowledge, been benchmarked systematically on these data, and the effect of a city-level hold-out on the estimated accuracy has not been measured.

Problem statement. Each case consists of a building-height raster $H(x, y)$ on a doubly periodic domain and a prevailing wind direction θ. We study two tasks: at *case level*, predicting summary statistics of the normalised pedestrian-level wind over open ground; at *field level*, predicting the normalised wind speed at every open-ground cell. Inputs are derived from (H, θ) only, and the models must generalise to city and layout groups absent from training.

This paper reports the first phase of a two-phase study and is restricted to classical machine learning. Its contributions are:

- a data audit of all 538 cases, with three corrections that affect the targets (Section III);
- 34 case-level and 30 pixel-level morphometric features, including the Macdonald and Kanda roughness parameterisations and periodic fetch and sheltering features (Section IV);
- a comparison of a mean predictor and seven regressors under nested grouped cross-validation, with group-bootstrap intervals, Shapley additive explanations (SHAP) and the measured optimism of random splits (Sections V and VI);
- a pixel-level model with diagnostics that locate the limit of tabular field prediction (Section VII), and an evaluation on a frozen test set (Section VIII).

The remainder of the paper is organised as follows. Section II reviews related work. Section III describes the data, its audit and the evaluation splits, and Section IV the features. Section V sets out the validation protocol, the models and the metrics. Sections VI and VII present case-level and field-level results, and Section VIII the evaluation on the frozen test set. Section IX discusses the results and their limitations, and Section X concludes. Fig. 1 gives an overview of the pipeline.

II. Related Work

Canopy aerodynamics and morphometry. Urban meteorology summarises a neighbourhood by plan-area density λ_p, frontal-area density λ_f and height statistics, and maps them to the zero-plane displacement height d and roughness

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Code, data tables and all reported results are available at https://github.com/mahir-m01/Urban-Microclimate-Modelling.

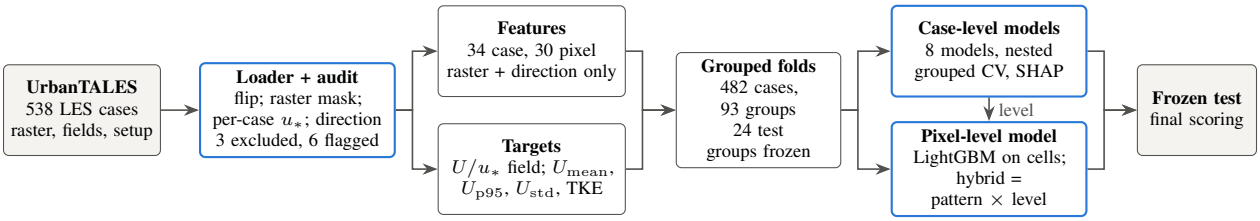


Fig. 1. Phase 1 pipeline. The loader applies every data correction once; features never read a simulated field; all model selection happens inside grouped folds of the development cases; the frozen test groups are never used for training or model selection and are scored only at the end.

length z_0 of the logarithmic wind profile above the canopy [5]. Macdonald et al. [6] derived d and z_0 for uniform arrays from a drag balance, and Kanda et al. [7] fitted corrections for height variability to LES of real city blocks; the Urban Multi-scale Environmental Predictor (UMEP) implements both [8]. Duan and Takemi [9] predicted d and z_0 with random forests. Within the canopy, Bentham and Britter [10] derived a spatially averaged velocity $U_c/u_* = (2/\lambda_f)^{1/2}$ from a momentum balance, and Coceal and Belcher [11] developed a canopy model of the mean wind profile. These parameterisations describe flow averaged over or above the canopy; we test them as features for the flow at pedestrian height.

Flow surrogates. Guo et al. [12] first used convolutional networks to approximate steady flow around shapes, Farimani et al. [13] trained conditional generative adversarial networks [14] to reproduce steady heat-conduction and fluid-flow solutions, and U-Nets predict Reynolds-averaged airfoil flows with errors of a few percent [15]. For pedestrian wind, Lu et al. [4] trained a U-Net on 512 UrbanTALES cases and report R^2 of about 0.94 on idealized arrays and 0.65 on the full dataset, with a random 80/20 case split. Snaiki et al. [16] refine U-Net predictions with a conditional generative adversarial network, and Huang et al. [17] stitch patch predictions with an inpainting U-Net, reporting R^2 of 0.66 and 0.69 on two representative held-out realistic cases. Clemente et al. [18] trade accuracy against speed in configurable convolutional networks. Lu et al. [19] proposed geometric parameters that distinguish flow over realistic layouts from flow over idealized arrays, which is why both subsets are needed.

Evaluation with dependent samples. When samples share structure, random cross-validation overstates generalisation, and held-out groups are the remedy [20]; spatial validation can reduce apparent accuracy drastically [21]. Selecting hyperparameters on the folds used for scoring biases the estimate as well [22], [23], and information unavailable at prediction time can leak into scores in subtler ways [24]. Neighbourhoods of one city share street patterns, and an idealized layout family is simulated at two wind directions, so both group dependence and selection bias must be controlled here.

III. Data

A. The UrbanTALES simulations

Each case is an LES with the PALM model system [25] of neutral flow over a doubly periodic domain 128 m high, driven by a constant horizontal pressure gradient that is identical across cases except one, which is within 2% (Table I). Sub-grid turbulence follows a 1.5-order closure [26]. The released fields are time averages over the windows in Table I on the horizontal slice at $z = 1.75$ m: the wind speed U (the time average of the instantaneous speed), the velocity components, the turbulent kinetic energy (TKE) and the shear stress. Each case also provides a building-height raster, horizontally averaged profiles and the run configuration. We keep the dataset's names for its two subsets, idealized and realistic. Fig. 2 shows three cases.

TABLE I
The two UrbanTALES subsets

	Idealized	Realistic
Cases	224	314
Layouts	aligned and staggered arrays	neighbourhoods in 22 countries, 50 city codes
λ_p	0.0625–0.64 (8 levels)	0.06–0.54
Heights	mean 16 m, σ_H 0–5.8 m	OpenStreetMap or uniform
Wind directions	0°, 45°	mostly 0° or 90°; 67 cases at others
Grid spacing	0.5 m	1 m (0.5 m for 5 cases)
Averaging window	3–11 h	4–12 h (4–9 h for 22 cases)[a]

[a] JP-Tok-V4 stopped early and averages 0.9 h (Section III-C).

B. Normalisation by each case's friction velocity

The cases share the pressure gradient Π, but the air volume differs between them because buildings displace air. The top boundary is free-slip, so in steady state the pressure force on the air is balanced by the drag on the ground and the buildings, $\rho u_*^2 A = \Pi V_{\mathrm{air}}$, where A is the horizontal domain area; the friction velocity u_* therefore grows with the air volume. The metadata values of u_* follow this balance exactly, and u_* ranges from 0.2015 to 0.2106 m s^{-1}. The domains are periodic and have no inlet, so the inlet speed often used to normalise CFD fields does not exist here; u_* is the velocity scale of this pressure-driven flow. We therefore model U/u_* with each case's own u_*, taken from the official metadata through a name map: 148 cases are named differently in the portal and the metadata, and the three that cannot be matched are the three excluded cases (Section III-C). Because u_* follows from the geometry through this balance, normalising by it uses no information from the simulated wind, and a prediction for a new layout converts back to m s^{-1} with u_* computed from its air volume.

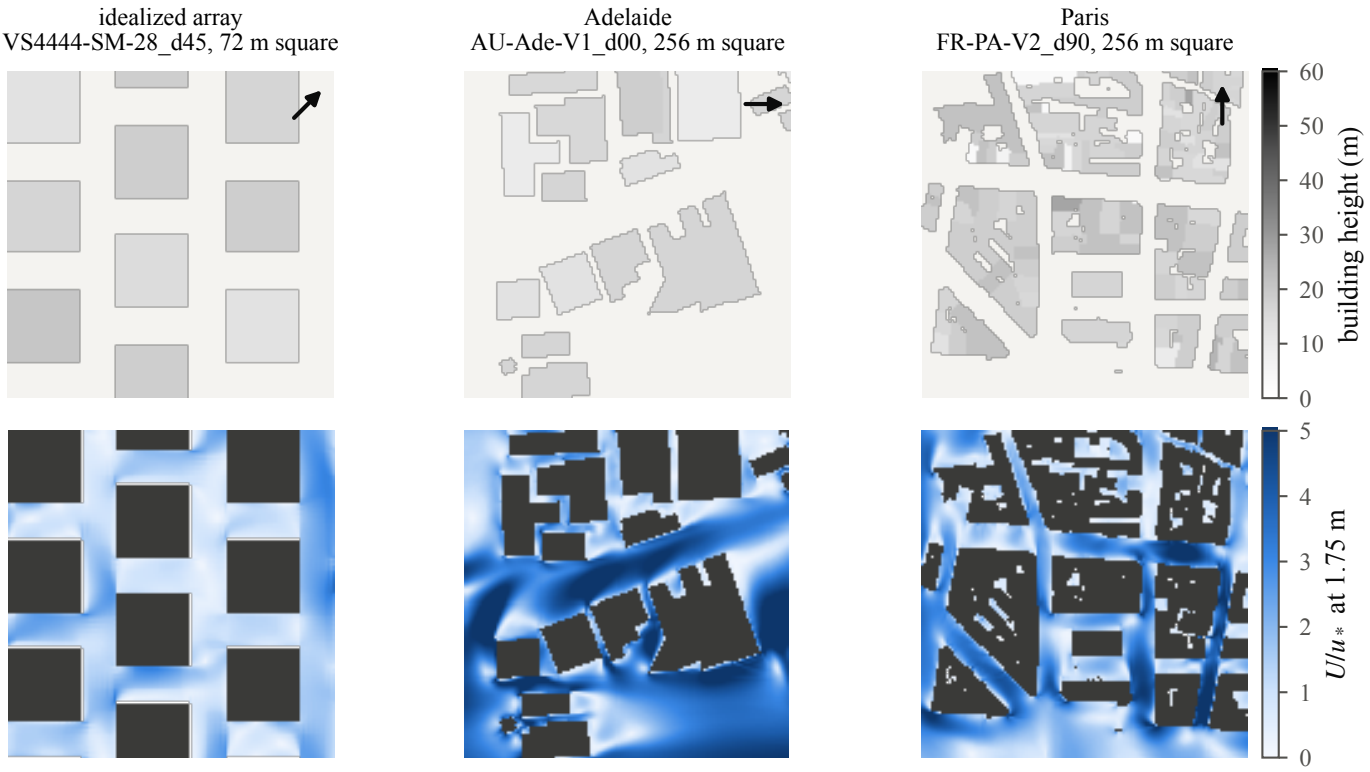


Fig. 2. Three development cases. Top: building heights; the arrow shows the direction the wind blows towards. Bottom: LES wind speed at pedestrian height, normalised by the case's friction velocity. Buildings are dark grey. Each panel shows at most 256 by 256 cells from a corner of the domain. Jets form where streets align with the wind; wakes and recirculation fill the spaces behind buildings.

C. Audit and corrections

Every file of all 538 cases was opened and checked for internal consistency (Table II, Fig. 3). Three findings affect the targets:

1) *Orientation.* The rasters are written top row first. Open ground in the raster coincides with the cells that hold a wind value on a median 0.71 of cells as stored, and 0.99 after a vertical flip.
2) *Masking.* Some files store zeros or interpolated values inside buildings, so a mask taken from missing values in the wind field leaks building cells into the average. In addition, 284 of the 292 realistic development cases contain open-ground cells with a speed of exactly zero (up to 6.7% of their open ground), which the simulation treated as buildings. Taking open ground from the raster ($H = 0$) and leaving out these cells changes the mean wind of 59 development cases by more than 1%, and by up to 14.6%. They include all 6 flagged cases, most of whose rasters also disagree with their wind fields.
3) *Scale.* The friction velocity varies by 4.5% across cases (Section III-B); normalising by a single nominal value would shift the normalised wind by up to that amount between cases.

A single loader applies the flip, the raster mask, the per-case u_* and the wind direction computed from the pressure-gradient vector of each run, so that every stage sees the same corrected data. Three cases whose downloadable rasters do not match the simulated domain are excluded. Six cases are kept but flagged: one run stopped early, and five rasters disagree with their wind fields on up to 13% of building cells; case- and field-level development scores are also computed without them. The metadata frontal-area density of realistic cases is 1.4 to 2 times the value computed from the raster, so the raster value is used throughout, consistent with the other features.

TABLE II
DATA AUDIT: FINDINGS AND HANDLING

Finding	Handling
Raster rows reversed against wind fields	flip in the loader
Values inside buildings; exact zeros on open ground (59 cases biased $>1\%$)	mask from raster, $H = 0$, without zeros
Per-case u_* from 0.2015 to 0.2106 m s^{-1}	normalise by own u_*
148 names differ between portal and metadata	name map; 3 unmatched
CN-BE-V1 names offset 90° from forcing	direction from ∇p
3 rasters do not match the simulated domain	excluded
JP-Tok-V4 stopped at 4.9 h (0.9 h average)	flagged
5 rasters disagree on $\leq 13\%$ of building cells	flagged
Metadata λ_f 1.4–2× raster value (realistic)	raster value used

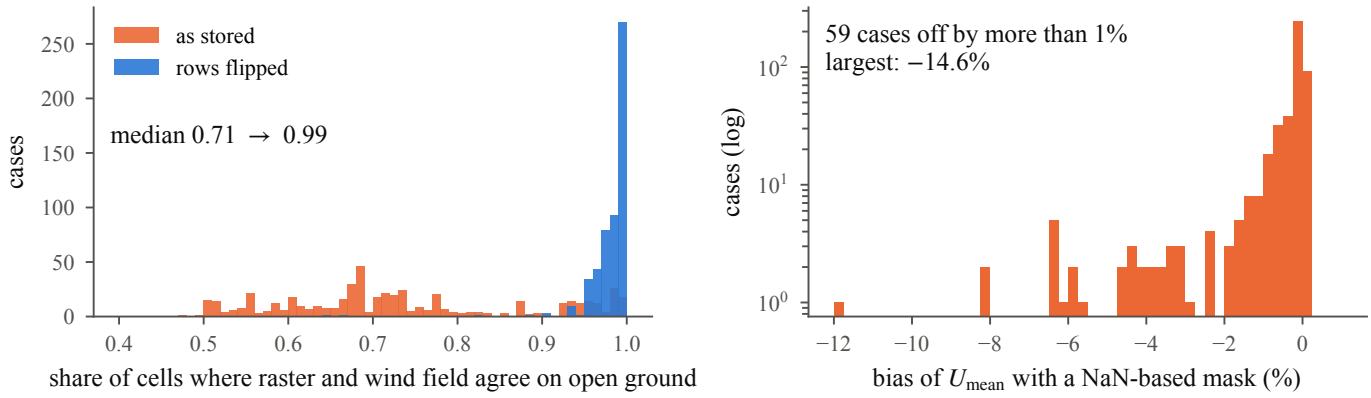


Fig. 3. Two data issues. Left: share of cells on which open ground in the building raster ($H = 0$) coincides with cells that hold a wind value, before and after flipping the raster rows, over all cases. Right: bias in the mean wind of each development case when open ground is taken from missing values in the wind field instead of from the raster without exact-zero cells (log scale).

D. Targets

For each case the four regression targets are statistics of U_i/u_* over the set Ω of open-ground cells:

$$U_{\mathrm{mean}} = \frac{1}{|\Omega|}\sum_{i\in\Omega}\frac{U_i}{u_*}, \qquad U_{\mathrm{p95}} = Q_{0.95,\, i\in\Omega}\left(\frac{U_i}{u_*}\right), \quad (1)$$

the standard deviation U_{std} of U_i/u_* over Ω, and the mean of TKE_i/u_*^2 over Ω. U_{p95} is the speed exceeded on 5% of open ground, a spatial analogue of exceedance-based comfort criteria. At field level the target is U_i/u_* at every open cell. In the idealized files the open cells directly north and east of each building carry no value (median 2.1%, at most 11.1% of open ground) and are left out.

E. Exploratory analysis

Fig. 4 shows U_{mean} on the 482 development cases. Its mean is 1.86; realistic cases are windier on average than idealized arrays, with a few very open neighbourhoods above 5. U_{mean} decreases with both λ_p and $\lambda_f(\theta)$, most clearly on idealized arrays, where each density level forms a tight cluster. Realistic cases scatter widely at a given density, so density alone does not determine the wind; this motivates features that describe openness and exposure (Section IV). The two subsets also occupy different ranges of λ_f, which anticipates the limited transfer between them (Section VI-C).

Table III summarises the four targets. Realistic cases have higher means of every speed statistic, and wider spreads of U_{p95} and U_{std}. U_{p95} and U_{std} are strongly correlated with U_{mean} across cases (Spearman ρ of 0.92 and 0.87) and TKE less so (0.71), so the targets are related but not redundant. Group sizes are uneven, from one case to 39 (median 4), which is why the bootstrap resamples groups rather than cases (Section VI-B). Among the 34 features, 8 pairs have $|r| > 0.95$, for example $\sigma_H/\bar{H}$ and γ_H, which are equal by construction, and λ_p and Macdonald's $d/\bar{H}$; this motivates the correlation filter in Section VI and the caveat on SHAP.

F. Splits

Cases are grouped so that near-duplicates stay together: realistic cases by city, so that every neighbourhood of one city, including the 0.5 m copies of Melbourne layouts, falls in one group, and idealized cases by layout family, so that both wind directions and both height spreads of a layout fall in one group. A frozen test set of 24 groups (53 cases) was fixed before modelling, with groups then defined by city code: seven realistic city groups stratified by density class, and 17 idealized families at 7 of the eight density levels, all of which also occur in development, so the idealized test cases test interpolation within the training range. The test groups are never used for training or model selection (Section VIII). The remaining **482 development cases** form 93 groups, assigned to five fixed folds of similar size by grouped k-fold, so that each fold holds out whole cities and layout families. The folds are not stratified: each holds 34 to 44 idealized cases, and the mix of density classes varies, but every class occurs in every training set, and scores are pooled over all out-of-fold predictions. Because the test groups were defined by city code, two test cases come from cities that also have development cases (Melbourne and Helsinki), and 11 idealized test families share their footprint with a development family of another height spread; test scores are also reported without the shared-city cases.

TABLE III
DEVELOPMENT TARGETS: MEAN (STANDARD DEVIATION) PER SUBSET, AND SPEARMAN CORRELATION WITH U_{mean} OVER ALL 482 DEVELOPMENT CASES

Target	Idealized	Realistic	ρ with U_{mean}
U_{mean}	1.64 (0.67)	2.00 (0.59)	1.00
U_{p95}	2.92 (0.82)	4.06 (0.97)	0.92
U_{std}	0.78 (0.23)	1.13 (0.28)	0.87
TKE/u_*^2	1.76 (0.71)	1.83 (0.49)	0.71

IV. Features

All features are computed from the building raster and the wind direction only; no feature reads a simulated field. Domains are periodic, so every distance, convolution and ray wraps around the edges. Street orientation enters through the facet-orientation entropy and the angle between the wind and the dominant facet orientation. The wind direction is the

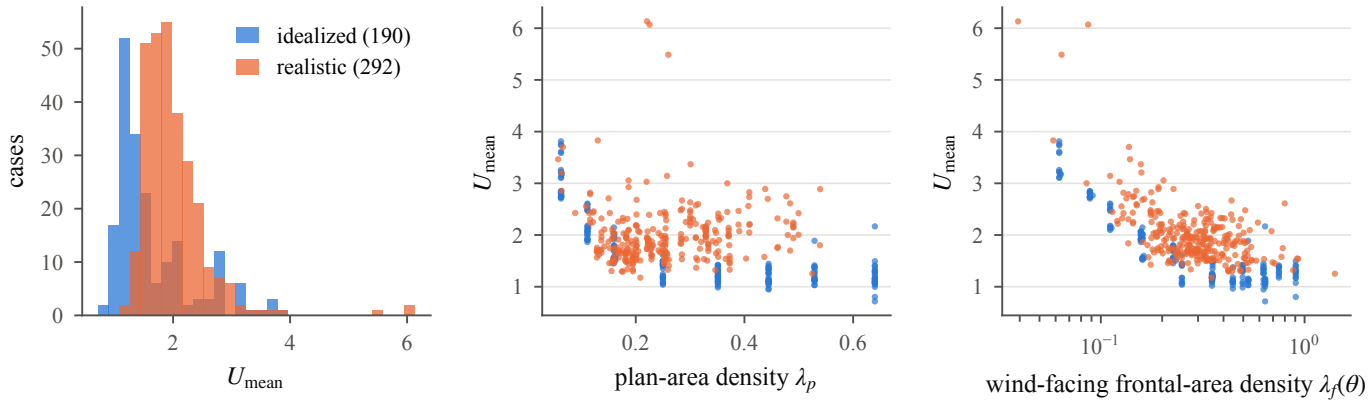


Fig. 4. Exploratory analysis of the 482 development cases. Left: distribution of U_{mean} per subset. Centre and right: U_{mean} against plan-area density and wind-facing frontal-area density. Idealized arrays form tight clusters per density level; realistic neighbourhoods scatter widely at a given density.

TABLE IV
THE 34 CASE-LEVEL FEATURES IN THREE NESTED SETS

Set	n	Features
Basic	9	λ_p; $\bar{H}$; σ_H; $H_{\max}$; $\lambda_f(\theta)$; wind direction θ, $\cos\theta$, $\sin\theta$; subset flag
Morphology	+8	wall-area density λ_w; height skewness; $\sigma_H/\bar{H}$, also as γ_H in %; $H_{\max}/\bar{H}$; built volume per unit area; buildings per hectare; mean footprint area
Physics	+17	$\lambda_{f,x}$, $\lambda_{f,y}$; Macdonald $d/\bar{H}$, $z_0/\bar{H}$; Kanda $d/H_{\max}$, $z_0/H_{\max}$; upwind fetch (mean, median); downwind and lateral fetch (means); share of open cells with upwind fetch above 50 m; distance to the nearest wall (mean, median, 90th percentile); facet orientation entropy; windward wall share; wind angle relative to the dominant facet orientation

only meteorological input that varies, because every run is neutral and driven by practically the same pressure gradient (Section III).

A. Case-level features

The 34 case-level features fall into three nested sets (Table IV).

Frontal area. Facets facing $+x$ and $+y$ are summed from height steps on the raster, giving $\lambda_{f,x}$ and $\lambda_{f,y}$. For a wind towards θ, $\lambda_f(\theta) = |\cos\theta|\,\lambda_{f,x} + |\sin\theta|\,\lambda_{f,y}$.

Macdonald. For an array of mean height $\bar{H}$ [6],

$$\frac{d}{\bar{H}} = 1 + \alpha^{-\lambda_p}(\lambda_p - 1), \tag{2}$$

$$\frac{z_0}{\bar{H}} = \left(1 - \frac{d}{\bar{H}}\right)\exp\left[-\left(\frac{\beta C_D}{2\kappa^2}\left(1 - \frac{d}{\bar{H}}\right)\lambda_f\right)^{-1/2}\right], \tag{3}$$

with $\lambda_f = \lambda_f(\theta)$, Macdonald's staggered-array values $\alpha = 4.43$ and $\beta = 1$, $C_D = 1.2$ and $\kappa = 0.4$. The exponential term follows from the drag balance on the obstacles. At fixed λ_p, z_0 rises with λ_f; as the packing density (λ_p and λ_f together) increases, it first rises and then falls as d approaches $\bar{H}$.

Kanda. For variable heights [7], with $X = (\sigma_H + \bar{H})/H_{\max}$ and $Y = \lambda_p\sigma_H/\bar{H}$,

$$\frac{d}{H_{\max}} = c_0X^2 + (a_0\lambda_p^{b_0} - c_0)X, \tag{4}$$

$$z_0 = (b_1Y^2 + c_1Y + a_1)\,z_{0,\mathrm{Mac}}, \tag{5}$$

with $a_0 = 1.29$, $b_0 = 0.36$, $c_0 = -0.17$, $a_1 = 0.71$, $b_1 = 20.21$ and $c_1 = -0.77$, as published and as in UMEP [8]. Uniform heights give $X = 1$ and $Y = 0$, so $d = a_0\lambda_p^{b_0}\bar{H}$ and $z_0 = 0.71\,z_{0,\mathrm{Mac}}$. Because d cannot exceed the tallest roof, it is capped at $0.999\,H_{\max}$ for 23 dense cases ($\lambda_p \geq 0.52$), where the fit leaves its range.

Fetch and openness. A periodic ray march from every open cell finds the distance to the first building upwind (capped at 150 m), downwind and to either side (each capped at 60 m). An exact Euclidean distance transform on the torus gives the distance to the nearest wall. Case-level features are their means and quantiles over open ground.

B. Pixel-level features

Each open cell receives 30 features (Fig. 5):

- local λ_p and mean height within radii of 8, 16, 32 and 64 m (8 features), computed by circular convolution in Fourier space;
- the distance to the nearest wall;
- upwind, downwind, left and right lateral fetch;
- the height of the tallest building upwind within 10, 10 to 30 and 30 to 60 m;
- canyon width (twice the wall distance) and aspect ratio;
- wind direction ($\cos\theta$, $\sin\theta$);
- six case descriptors (λ_p, $\bar{H}$, σ_H, $H_{\max}$, $\lambda_f(\theta)$, subset flag);
- four sheltering features.

The sheltering features encode the fact that a building of height h shelters a wake a few heights long. They are the largest ratio h/x over the three upwind distance bands (x the band centre), the upwind fetch along rays rotated by $\pm30°$ from the wind (two features), and the upwind fetch divided by $\bar{H}$.

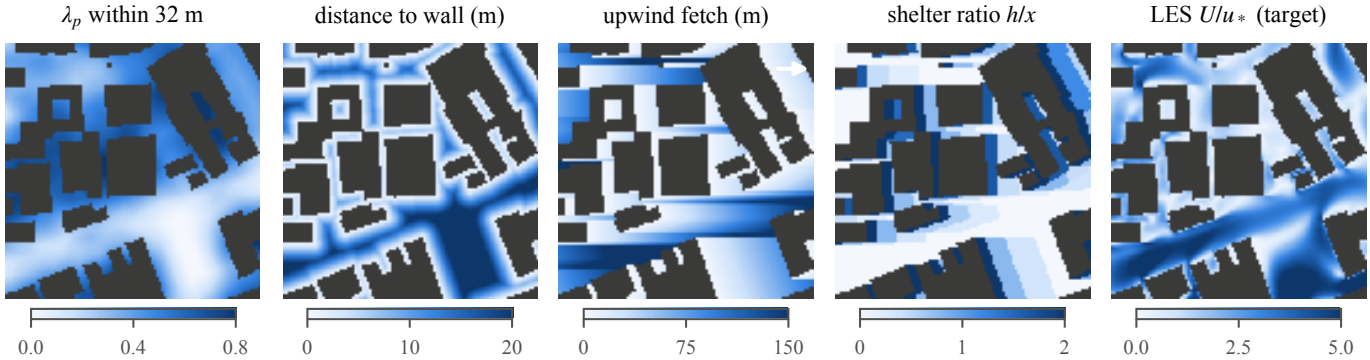


Fig. 5. Four pixel features and the target on part of a realistic case (AU-Ade-V1, wind towards $+x$). Local density and distance to walls describe openness; upwind fetch and the shelter ratio h/x describe exposure to the oncoming wind.

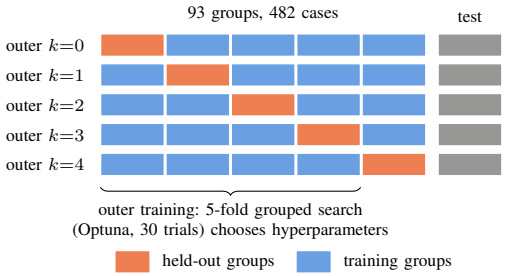


Fig. 6. Nested grouped cross-validation. Each outer fold holds out whole city and layout groups (orange); hyperparameters are chosen by a grouped search inside the training folds only. The frozen test groups (grey) are not used until the final scoring.

V. Methodology

A. Nested grouped cross-validation

The outer loop runs over the five fixed development folds, each of which holds out whole city and layout groups (Fig. 6). Inside each outer-training set, a five-fold grouped split on the same groups scores candidate hyperparameters: Optuna [27] runs 30 trials of the tree-structured Parzen estimator (TPE) [28]. The best setting is refitted on the outer-training cases and scored on the outer fold. Feature scaling and target standardisation are fitted inside these loops. All models are implemented with scikit-learn [29], XGBoost and LightGBM, with random seed 42 throughout.

B. Case-level models

Eight models span the bias–variance range: the training mean; ridge regression; k-nearest neighbours; support-vector regression (SVR) with a radial basis function (RBF) kernel; a Gaussian process (GP) with a Matérn-5/2 kernel, one length scale per feature (automatic relevance determination [30]) and a white-noise term, whose hyperparameters maximise the marginal likelihood; random forests [31]; XGBoost [32]; and LightGBM [33]. On standardised features $\boldsymbol{x}$, ridge regression minimises $\sum_i (y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)^2 + \alpha\|\boldsymbol{w}\|^2$; k-NN averages the targets of the k nearest training cases; SVR fits $f(\boldsymbol{x}) = \sum_i (\eta_i - \eta_i^*) K(\boldsymbol{x}_i, \boldsymbol{x}) + b$ with $K(\boldsymbol{x}, \boldsymbol{x}') = \exp(-\gamma\|\boldsymbol{x} - \boldsymbol{x}'\|^2)$ under a loss that ignores errors smaller than ϵ; the GP predicts the posterior mean $\boldsymbol{k}_*^\top (\boldsymbol{K} + \sigma_n^2 \boldsymbol{I})^{-1} \boldsymbol{y}$, where $\boldsymbol{K}$ holds the kernel between training cases and $\boldsymbol{k}_*$ between them and the new case; and a random forest averages trees grown on bootstrap samples with random feature subsets, which reduces the variance of deep trees while keeping their low bias. Table V lists the search spaces. The two boosting libraries minimise the same regularised objective

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_t \left(\gamma_\mathrm{b} T_t + \tfrac{1}{2} \lambda_\mathrm{b} \|\boldsymbol{w}_t\|^2 \right), \qquad (6)$$

over trees t with T_t leaves and leaf values $\boldsymbol{w}_t$, where γ_b and λ_b penalise the number of leaves and their size. They differ in tree growth: XGBoost grows each tree level by level to a depth limit, whereas LightGBM grows it leaf by leaf on histogram-binned features. For a fair comparison both are tuned over the same search space, with an additional leaf-count range for LightGBM; the minimum leaf size maps to XGBoost's minimum child weight, which equals a sample count under squared loss. The reference *B1* is a random forest with default settings on the nine basic features, close to the inputs of earlier morphology-based baselines [4]. Default-setting versions of the three ensembles isolate the effect of tuning.

TABLE V
Hyperparameter search spaces (Optuna TPE, 30 trials, five grouped inner folds) and fixed settings. Ranges marked log are sampled on a log scale.

Model	Search space or setting
Ridge	$\alpha \in [10^{-3}, 10^{3}]$ (log)
k-NN	$k \in \{2, \ldots, 30\}$; uniform or distance weights
SVR	$C \in [0.1, 300]$, $\gamma \in [10^{-3}, 1]$, $\epsilon \in [0.01, 0.5]$ (all log)
GP	Matérn-5/2 ARD + white noise; marginal likelihood, 2 restarts
Random forest	500 trees; feature fraction $\in [0.2, 1]$; minimum leaf $\in \{1, \ldots, 10\}$; depth $\in \{8, 16, \text{none}\}$
XGBoost and LightGBM	trees $\in \{200, \ldots, 1500\}$; learning rate $\in [0.01, 0.2]$ (log); depth $\in \{2, \ldots, 8\}$; minimum leaf $\in \{1, \ldots, 40\}$ (log); row subsample $\in [0.6, 1]$; column subsample $\in [0.5, 1]$; L2 penalty $\in [10^{-3}, 10]$ (log); LightGBM also leaves $\in \{4, \ldots, 64\}$ (log)
Defaults	XGBoost, LightGBM: 600 trees, learning rate 0.05 (LightGBM 31 leaves); random forest: 500 trees
Pixel model	LightGBM, fixed: 800 trees, learning rate 0.05, 127 leaves, minimum leaf 100, row and column subsample 0.8, L2 penalty 1

C. Significance, attribution and protocol checks

Fold scores are too few for a reliable test, and cases within a city group are correlated. We therefore use a paired percentile bootstrap [34] over groups: the 93 groups are resampled with replacement 2,000 times, the same resample is applied to the out-of-fold predictions of both models, and the 95% interval of ΔR^2 is reported. The fitted models are held fixed, so the interval reflects case sampling but not training variance, and no correction is applied for the nine comparisons. TreeSHAP [35], [36] attributes each prediction of the more accurate boosting model to the features, $f(\boldsymbol{x}) = \phi_0 + \sum_j \phi_j(\boldsymbol{x})$, where ϕ_0 is the mean prediction and the Shapley value ϕ_j is the change in prediction from adding feature j, averaged over all orders in which the features can be added. It uses each outer-fold model on its own held-out cases only. Two further checks measure how much the protocol matters: the same untuned models under random five-fold cross-validation, and transfer between the idealized and realistic subsets.

D. Pixel-level model and hybrid

The field model is LightGBM with the fixed setting of Table V on the 30 pixel features. It is trained on 3,000 random open cells per training case, so that every case carries equal weight (about 1.16 M cells per training fold), and it predicts every open cell of each held-out case. The *hybrid* predicts the pattern $(U_i/u_*)/U_{\mathrm{mean}}$ instead and multiplies it by the out-of-fold prediction of U_{mean} from the most accurate case-level model, LightGBM. Two level-only references assign a single value to every cell: the case-level prediction, and the true case mean (an oracle).

E. Metrics

At case level, R^2 and the mean absolute error (MAE) are computed on the pooled out-of-fold predictions of all development cases, overall and per subset. At field level, with errors $e_i = \hat{y}_i - y_i$ over the open cells Ω_c of case c,

$$R^2_{\mathrm{pool}} = 1 - \frac{\sum_c \sum_{i\in\Omega_c} e_i^2}{\sum_c \sum_{i\in\Omega_c} (y_i - \bar{y})^2}, \tag{7}$$

$$R^2_c = 1 - \frac{\sum_{i\in\Omega_c} e_i^2}{\sum_{i\in\Omega_c} (y_i - \bar{y}_c)^2}, \tag{8}$$

where $\bar{y}$ is the mean over all cells and $\bar{y}_c$ the mean over case c. R^2_{pool} weights cells equally, so larger domains count more and correct case levels are rewarded; R^2_c is normalised by the within-case variance and is therefore dominated by pattern error. We report the median of R^2_c over cases. The field MAE is the per-case MAE averaged over cases, and the normalised RMSE of a case is its RMSE divided by its mean true speed, $\mathrm{NRMSE}_c = \mathrm{RMSE}_c/\bar{y}_c$, reported as the median over cases. The per-case mean squared error splits into level and pattern,

$$\frac{1}{|\Omega_c|}\sum_{i\in\Omega_c} e_i^2 = \bar{e}_c^{\,2} + \frac{1}{|\Omega_c|}\sum_{i\in\Omega_c} (e_i - \bar{e}_c)^2, \tag{9}$$

with $\bar{e}_c$ the mean error of case c. A flat field at the right level has no level error, and its pattern error equals the variance of the true field. Finally, ΔU_{p95} is the mean over cases of the predicted minus the true 95th percentile of U_i/u_*; negative values mean underprediction.

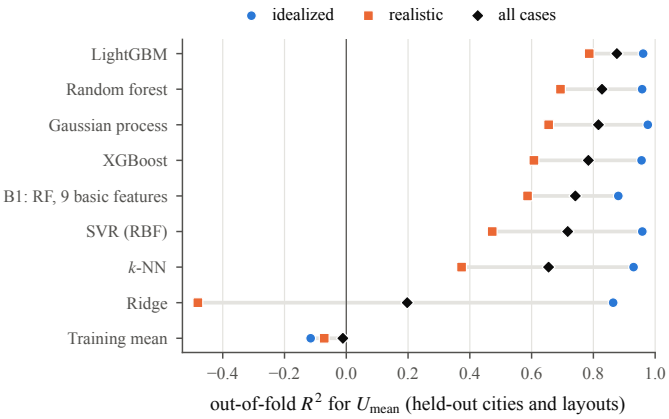


Fig. 7. Out-of-fold R^2 for U_{mean} on held-out city and layout groups, models sorted by R^2 over all cases. Realistic cases separate the models; idealized arrays are easy for most non-linear learners.

VI. Case-Level Results

A. Model comparison

Table VI and Fig. 7 give the out-of-fold scores.

Linear and kernel models. Ridge regression fails on real cities ($R^2 = -0.480$, against 0.864 on idealized arrays), consistent with a strongly non-linear relation between geometry and wind that differs between the two kinds of layout; on one fold its R^2 is -1.48, because a linear model extrapolates far outside the training range for a held-out city. k-NN, SVR and the GP recover part of the relation (realistic R^2 0.373, 0.472 and 0.655). The GP is the most accurate model on idealized arrays ($R^2 = 0.976$), consistent with smooth interpolation suiting that structured design space.

Tree ensembles. LightGBM is the most accurate model overall ($R^2 = 0.876$, realistic 0.787), followed by the random forest (0.828) and the GP (0.816); tuned XGBoost reaches 0.784. The MAE of LightGBM, 0.143, is about 8% of the mean development U_{mean}.

Other targets. High-percentile speeds are harder to predict than means. U_{p95} reaches $R^2 = 0.767$ overall but 0.571 on real cities, where they are likely set by individual corners and gaps that case-level summaries do not describe. TKE is the easiest target, with R^2 of 0.898 to 0.947 for the tree ensembles, SVR and the GP, consistent with canopy turbulence being set mainly by drag, which λ_f and z_0 summarise. Removing the six flagged cases changes R^2 by at most 0.007 for any model and target.

B. Statistical significance

Fig. 8 shows the group-bootstrap intervals.

Against the baseline. LightGBM and the random forest exceed B1 by 0.135 [0.095, 0.178] and 0.086 [0.039, 0.137] in R^2, and both intervals also exclude zero for U_{p95}. The gains of XGBoost, 0.042 [−0.028, 0.113], and of the GP, 0.075 [−0.042, 0.157], are not significant.

TABLE VI
CASE-LEVEL RESULTS: OUT-OF-FOLD SCORES ON 482 DEVELOPMENT CASES (190 IDEALIZED, 292 REALISTIC) ON HELD-OUT CITY AND LAYOUT GROUPS. R^2 AND MAE ARE POOLED OVER FOLDS; "FOLD SD" IS THE STANDARD DEVIATION OF R^2 ACROSS THE FIVE FOLDS. MAE OF U/u_* (DIMENSIONLESS). RF: RANDOM FOREST. BOLD: BEST VALUE PER COLUMN (HIGHEST R^2, LOWEST MAE).

	U_{mean}					U_{p95}	U_{std}	TKE
Model	R^2	R^2 ideal.	R^2 real.	MAE	fold SD	R^2	R^2	R^2
Training mean	−0.011	−0.115	−0.071	0.470	–	−0.006	−0.003	−0.019
Ridge	0.197	0.864	−0.480	0.344	0.936	0.380	0.374	0.595
k-NN	0.655	0.930	0.373	0.219	0.065	0.668	0.641	0.834
SVR (RBF)	0.717	0.958	0.472	0.187	0.065	0.700	0.687	0.906
Gaussian process	0.816	**0.976**	0.655	0.164	0.141	0.710	0.685	**0.947**
Random forest	0.828	0.958	0.694	0.158	0.025	0.725	0.683	0.898
LightGBM	**0.876**	0.961	**0.787**	**0.143**	0.033	**0.767**	0.694	0.925
XGBoost	0.784	0.956	0.608	0.163	0.051	0.723	**0.741**	0.904
B1: RF, basic, default	0.742	0.881	0.587	0.230	0.025	0.660	0.584	0.860
RF, default	0.831	0.955	0.703	0.155	0.029	0.746	0.705	0.899
LightGBM, default	0.829	0.948	0.704	0.158	0.029	0.723	0.682	0.909
XGBoost, default	0.851	0.947	0.749	0.153	0.035	0.742	0.708	0.884

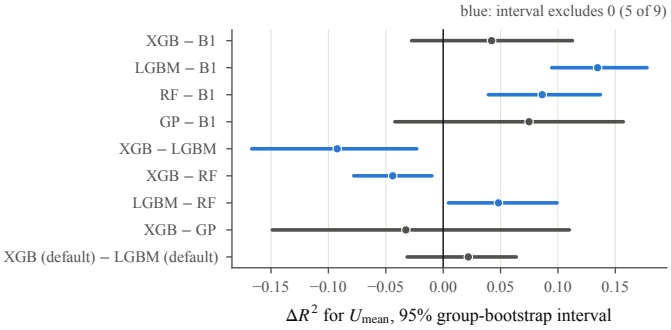


Fig. 8. Differences in U_{mean} R^2 with 95% paired intervals from 2,000 resamples of whole city and layout groups. XGB: XGBoost; LGBM: LightGBM; RF: random forest; GP: Gaussian process; B1: nine-feature baseline. Blue: the interval excludes zero.

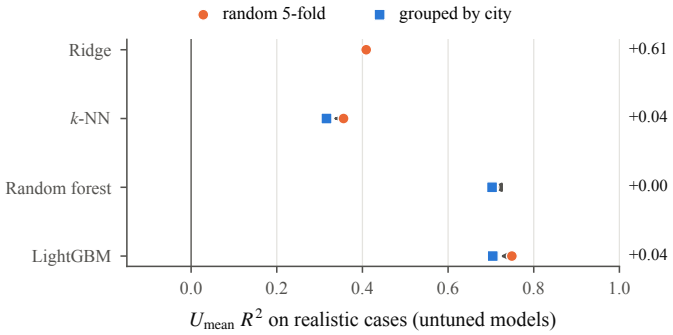


Fig. 9. U_{mean} R^2 on realistic cases under random five-fold cross-validation and under folds that hold out whole city groups, for the same untuned models. Numbers on the right are the optimism of the random split (random minus grouped).

Among the leading models. LightGBM exceeds the random forest by 0.048 [0.005, 0.099], and XGBoost−LightGBM is −0.092 [−0.167, −0.023]. With default settings, however, XGBoost−LightGBM is 0.022 [−0.031, 0.064], so the lead of tuned LightGBM reflects what each inner search found rather than a difference between the two boosting algorithms.

Effect of tuning. With default settings the three ensembles reach 0.831 (random forest), 0.829 (LightGBM) and 0.851 (XGBoost). Tuning raises LightGBM to 0.876 but lowers XGBoost to 0.784, and leaves the random forest almost unchanged. The loss of XGBoost lies almost entirely in one fold (R^2 0.727 tuned against 0.874 with defaults), where the setting chosen by the inner search strongly underpredicts two very open neighbourhoods, CA-Sunset and US-WestPhoenix ($U_{\mathrm{mean}} > 6.0$), whose cities are held out with them. Hyperparameters chosen on the training groups need not carry over to unseen cities, so tuning can lower held-out accuracy.

C. Sensitivity to the validation protocol

Random against grouped folds. On real cities, random five-fold cross-validation overstates R^2 for the same untuned models (Fig. 9). For tree ensembles the optimism is small: 0.00 for random forests and 0.04 for LightGBM, and 0.04 for k-NN. Ridge regression, which extrapolates poorly to unseen city groups, is strongly affected: its R^2 falls from 0.409 under random folds to −0.200 under grouped folds.

Transfer between subsets. Models do not transfer between the two subsets. Trained on idealized arrays only, LightGBM reaches $R^2 = 0.12$ on real cities; trained on real cities only, it reaches −0.51 on arrays. Both subsets must therefore be in the training data.

D. Feature relevance and attribution

Feature ablation. With the basic set, the tuned ensembles reach R^2 of 0.715 to 0.738 (realistic 0.505 to 0.552). Adding the morphology set raises XGBoost and LightGBM to 0.766 and 0.755 but lowers the random forest to 0.700. Adding the physics set raises the random forest to 0.828 and LightGBM to 0.876 (realistic 0.694 and 0.787), so for these two models most of the gain over the basic set comes from the physically motivated features; tuned XGBoost gains less (0.784).

Feature selection. A separate study, with LightGBM at fixed settings and selection fitted on the training folds only, found no clear gain from selection over the full set (pooled out-of-fold $R^2 = 0.814$). Correlation filtering keeps 28.0 features on average across folds and scores 0.819, recursive elimination

keeps 19.4 at 0.810, and principal components reach only 0.629 to 0.699, consistent with linear projections blurring the thresholds on which trees split.

SHAP attribution. TreeSHAP for LightGBM on held-out folds (Fig. 10) ranks the mean distance to walls and the wind-facing frontal area λ_f first, with nearly equal mean absolute contributions, followed by the 90th-percentile wall distance, upwind and lateral fetch, built volume, λ_p and the Macdonald roughness length. This order is consistent with canopy aerodynamics: frontal area sets the drag, and wall distance and fetch describe openness and exposure. The dependence panels are largely monotone with physically consistent signs: more open space and less frontal area raise the mean wind. The signs agree across subsets, but the subsets occupy different feature ranges, consistent with the poor transfer between them (Section VI-C). SHAP divides credit among correlated features (λ_f, z_0 and the distance statistics), so the ranking describes this model rather than a unique causal order.

VII. Field-Level Results

A. Level and pattern

Level only. A flat field at the true case mean has $\bar{e}_c = 0$ and pattern error equal to the within-case variance, so its R^2_{pool} equals the share of the pooled variance that lies between cases (law of total variance). It is 0.311: about a third of the variation at pedestrian height lies between neighbourhoods, and two thirds within them. The case-level prediction comes close to this oracle (0.270).

Pixel model. The pixel model reaches a pooled R^2 of 0.591 (Table VII), with median per-case R^2 of 0.418 on idealized and 0.247 on realistic cases. Its field MAE is 0.611 and its median NRMSE 0.468, against 0.816 and 0.549 for the oracle; the typical error is still about half of a case's mean speed, because the wind varies strongly within each neighbourhood. The sheltering features raise the realistic median per-case R^2 from 0.201 to 0.247, so their gain is in the pattern.

Hybrid. The hybrid performs like the direct model, because the pixel model already learns the level from its case descriptors.

Error decomposition. The pattern dominates the error (Fig. 11). High-percentile speeds are underpredicted ($\Delta U_{\mathrm{p95}} = -0.73$), and the absolute error grows from 0.64 in the cells next to walls (within 1 m) to 0.85 beyond 32 m from them, where speeds are highest. Next to walls the hybrid is no more accurate than the direct model (0.65 against 0.64).

B. Limits of the tabular representation

Fig. 12 shows held-out realistic cases. The model reproduces the broad structure: open squares are windy and narrow streets calm. It smooths corner jets, wakes and channelling, and its predictions show streaks along the wind direction that follow the ray-marched fetch features. In the worst case (AU-Mel-U15_d90), the wind blows across rows of buildings. Pedestrian-level flow there is likely dominated by canyon vortices in which the streamwise velocity reverses between streets, a structure that features computed along straight rays cannot encode.

Diagnostics. Two experiments on development cases separate the possible causes of this limit (Fig. 13). They were run after the test set had been scored, on development cases only, and changed no modelling choice:

1) *The split.* Under a random 80/20 case split, as in [4], the same model reaches a median per-case R^2 of 0.426 (idealized) and 0.264 (realistic), only slightly above grouped folds (0.418 and 0.247). The split does not limit the tree model.
2) *Capacity.* A far less regularised model (2,000 trees, 1,023 leaves, minimum leaf 5) raises the median per-case R^2 on 26 of its own realistic training cases from 0.373 to 0.586, but not on the held-out realistic cases of the random split (0.264 against 0.257). The additional capacity fits the training cases without generalising.

These results indicate that the 30 local features, rather than the split or the model capacity, limit the pattern that transfers to unseen neighbourhoods. Under a comparable, but not identical, random-split protocol (512 cases, different preprocessing), the published U-Net reports R^2 of about 0.94 on idealized arrays and 0.65 on the full set [4]. The same split gives a median per-case R^2 of 0.426 and 0.317 here, and a pooled R^2 of 0.450 on all cases; whichever aggregation the published values use, this suggests that most of the gap lies in the representation rather than the protocol.

VIII. Evaluation on the Frozen Test Set

After every modelling choice was fixed, the final models were re-tuned with the same inner search on all 482 development cases, refitted, and scored on the 53 cases of the 24 frozen test groups (Table VIII). The test set was first scored on 26 October 2026; an audit of the pipeline then corrected the features, the targets and the development groups, the models were re-tuned and re-selected on the development folds, and the test set was scored a second time on 27 October 2026 (UTC). No test score was used for any choice, and the scoring script records both runs and the reason for the second. Two test groups (AU-SURREYHILLS in Melbourne and FI-MS in Helsinki) come from cities that also have development groups, so scores are also given without them.

Generalisation. On the test set every non-linear model exceeds B1 for U_{mean} ($R^2 = 0.827$), including k-NN and SVR, which were below B1 in development (0.655 and 0.717 against 0.742). The field models reproduce the development results: pooled R^2 0.680, median per-case R^2 0.351 and median NRMSE 0.428 for the pixel model, against 0.378 and 0.518 for the oracle flat field.

Model ranking. The order differs from development: the GP ranks first ($R^2 = 0.931$, realistic 0.843), and LightGBM, the most accurate model in development, ranks 5th (0.865). On only 19 realistic test cases from seven city groups, a few cases decide such an order, so the test set confirms generalisation but cannot rank the leading models; the development comparison over 482 cases is the evidence for ranking. The all-case R^2 is also higher than in development because the test set is 64% idealized, against 39% in development.

Shared cities. The two shared-city test groups hold only two cases. Removing them changes case-level U_{mean} R^2 by

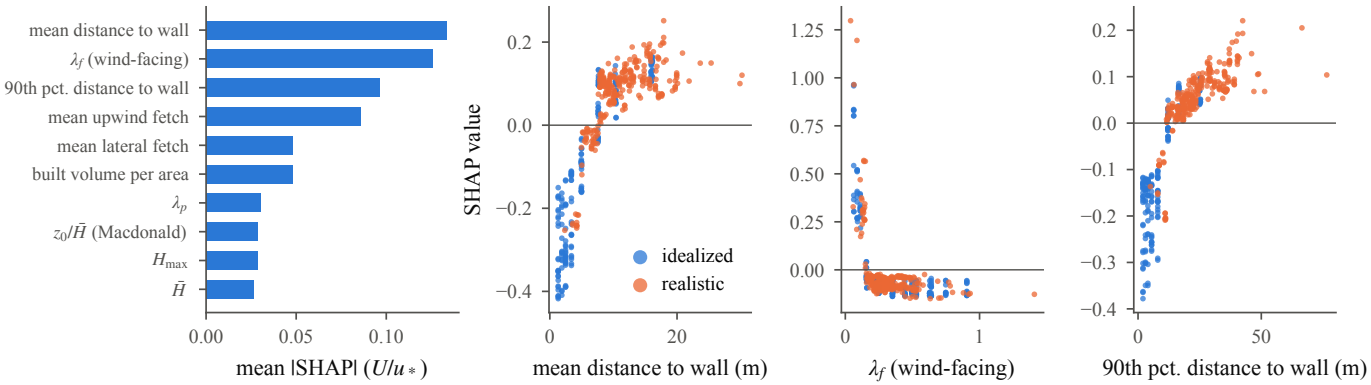


Fig. 10. TreeSHAP for LightGBM on held-out folds (U_{mean}). Left: mean absolute contribution of the ten leading features. Right: contributions of the three leading features against their values; more open space and less frontal area raise the mean wind.

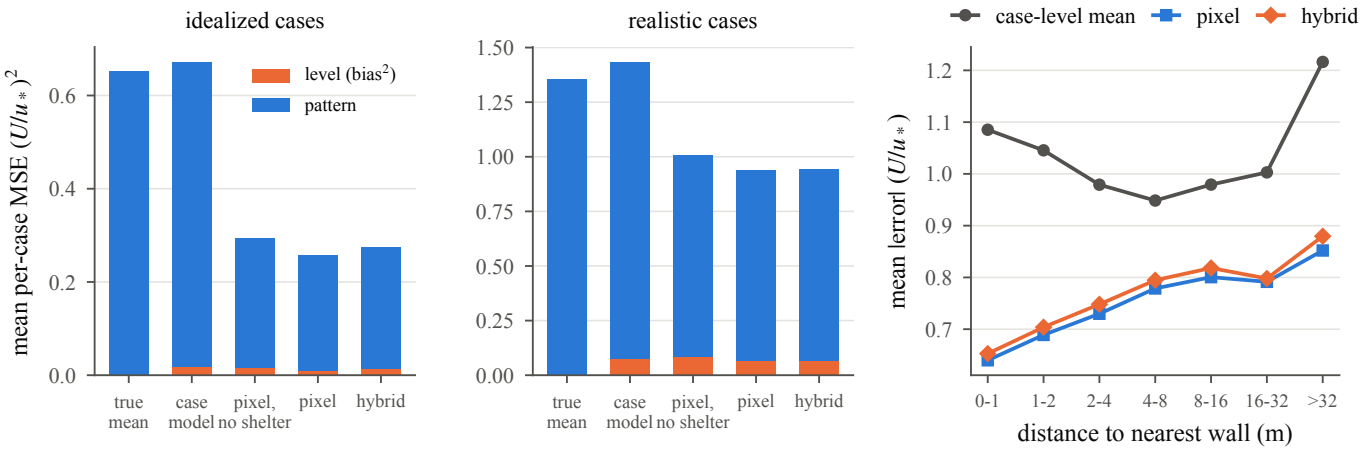


Fig. 11. Left and centre: per-case mean squared error split into level (bias^2) and pattern, for the level-only references and the learned models. Right: mean absolute error against distance to the nearest wall on realistic held-out cases.

TABLE VII

FIELD-LEVEL RESULTS ON HELD-OUT DEVELOPMENT CASES (190 IDEALIZED, 292 REALISTIC). POOLED R^2 IS COMPUTED OVER ALL OPEN CELLS AND R^2_c PER CASE; MAE IS AVERAGED OVER CASES; NRMSE IS THE PER-CASE RMSE DIVIDED BY THE CASE'S MEAN SPEED, AS A MEDIAN OVER CASES; ΔU_{p95} IS THE MEAN SHIFT OF THE 95TH PERCENTILE. ERRORS IN UNITS OF U/u_*. THE ORACLE'S PER-CASE R^2 IS ZERO BY CONSTRUCTION.

	pooled R^2			median R^2_c			MAE			median NRMSE			
Model	all	ideal.	real.	all	ideal.	real.	all	ideal.	real.	all	ideal.	real.	ΔU_{p95}
True mean (oracle)	0.311	0.394	0.298	0.000	0.000	0.000	0.816	0.641	0.929	0.549	0.479	0.566	−1.75
Case-level mean	0.270	0.390	0.252	−0.010	−0.003	−0.016	0.832	0.652	0.949	0.560	0.483	0.574	−1.76
Pixel, no sheltering	0.553	0.831	0.513	0.270	0.393	0.201	0.636	0.421	0.777	0.482	0.366	0.506	−0.77
Pixel	0.591	0.876	0.550	0.304	0.418	0.247	0.611	0.393	0.753	0.468	0.359	0.491	−0.73
Hybrid	0.576	0.866	0.534	0.320	0.431	0.251	0.615	0.401	0.755	0.467	0.357	0.492	−0.76

−0.03 to +0.02. Pooled field R^2 falls by 0.06 to 0.08 for every field model, the oracle included, which reflects the changed case mix, while the median per-case R^2 of the pixel model is unchanged (0.351); neither change points to leakage.

IX. DISCUSSION AND LIMITATIONS

Physical consistency. The leading features (frontal area, openness, fetch and roughness) are the quantities of canopy drag theory, and their SHAP effects have the expected signs in both subsets. Most of the gain over the nine-feature baseline comes from these physically motivated features rather than from tuning.

Practical accuracy. At case level the best model errs by about 8% of the mean wind on unseen groups (4% on idealized arrays, 9% on realistic cases), but realistic R^2 (0.787) remains below idealized (0.961), and high-percentile speeds on real cities reach only $R^2 = 0.571$. At field level about two thirds of the variance at pedestrian height lies within neighbourhoods; tabular models capture its level and broad structure, but not jets, wakes or canyon vortices, whose shape depends on the exact geometry around each point.

Limitations and threats to validity.

- The test groups were fixed by city code before the de-

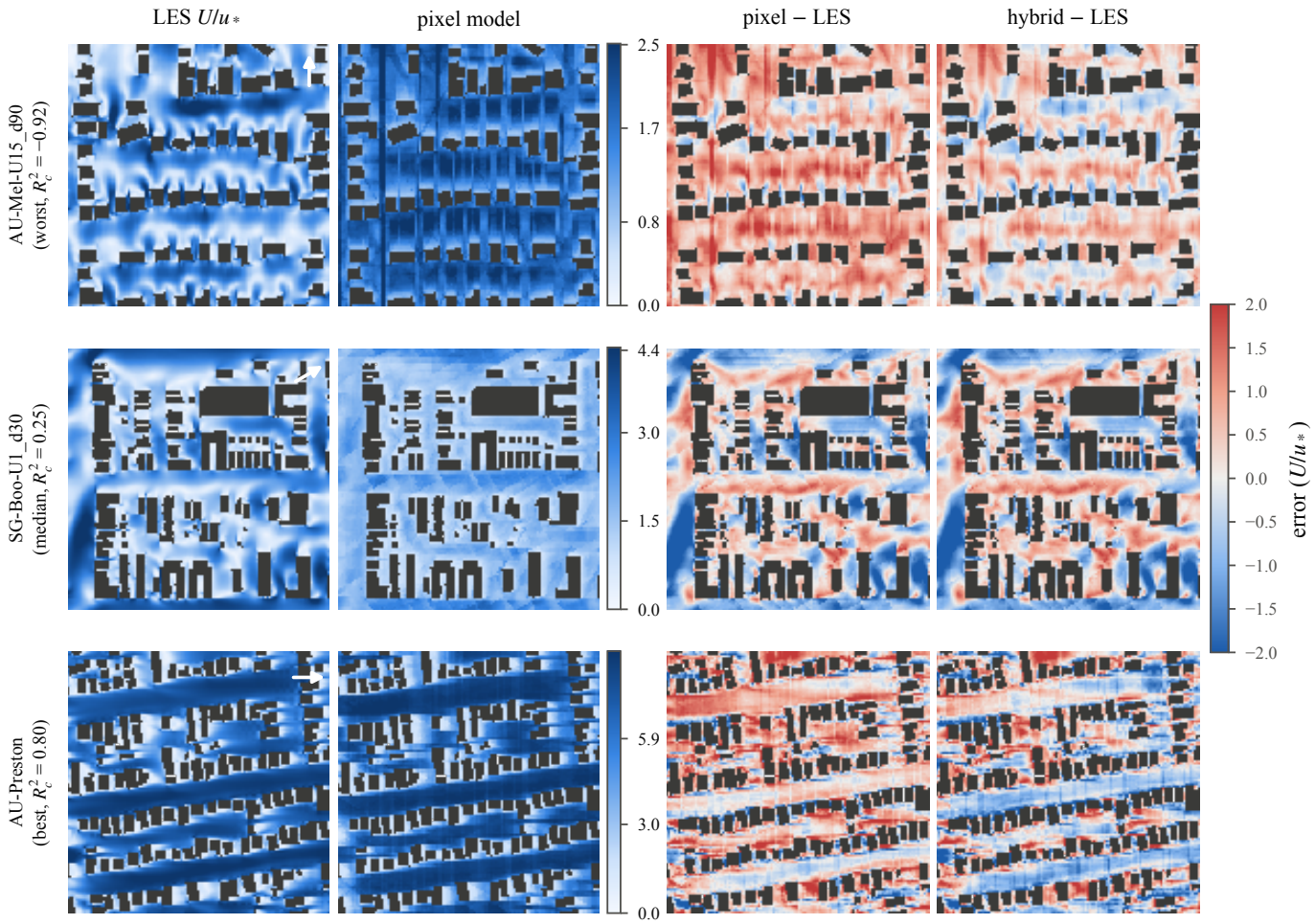


Fig. 12. Held-out realistic cases with the lowest, median and highest per-case R^2 (R_c^2) of the pixel model: LES, prediction, and errors of the pixel model and the hybrid (blue: too slow, red: too fast). Panels show at most 320 by 320 cells from a corner of the domain; R_c^2 is computed over the whole case. The model captures openness but smooths jets and wakes and misses high speeds.

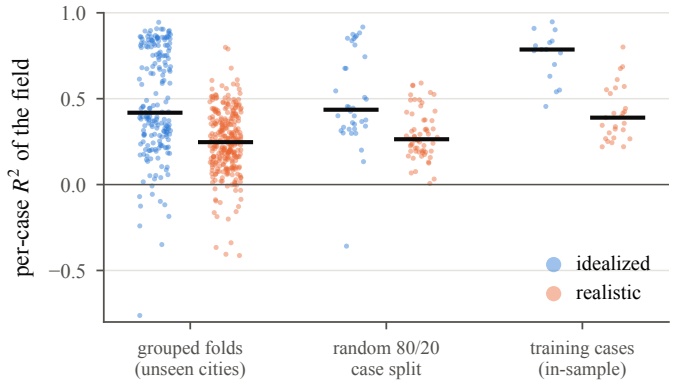


Fig. 13. Per-case R^2 of the field for the fixed-setting pixel model under three protocols (bars: medians). A random case split barely helps the tree model; on its own training cases it fits idealized arrays well but realistic neighbourhoods only moderately.

TABLE VIII
FROZEN TEST SET (53 CASES, 24 GROUPS)

	case level: U_{mean} R^2			
Case-level model	all	ideal.	real.	w/o shared city
Training mean	−0.013	−0.140	−0.074	−0.038
Ridge	0.724	0.840	0.507	0.747
k-NN	0.870	0.959	0.723	0.843
SVR (RBF)	0.857	0.974	0.669	0.865
Gaussian process	0.931	0.985	0.843	0.946
Random forest	0.883	0.971	0.740	0.861
LightGBM	0.865	0.974	0.690	0.875
XGBoost	0.898	0.975	0.772	0.892
B1	0.827	0.919	0.666	0.804
Field model	pooled R^2	med. R_c^2	MAE	NRMSE
True mean (oracle)	0.378	0.000	0.739	0.518
Case-level mean	0.295	−0.007	0.768	0.524
Pixel	0.680	0.351	0.530	0.428
Hybrid	0.685	0.366	0.532	0.427

velopment groups were merged by city, so two test cases share a city with development cases and 11 idealized test families share a footprint with a development family.

- Each model is fitted with one seed on five folds, and the bootstrap intervals do not include training variance.
- Uncertainty is quantified for the evaluation, through fold standard deviations and group-bootstrap intervals, but not for individual predictions; predictive intervals are deferred to Phase 2.

- The test set holds only 19 realistic cases from seven city groups, so it confirms generalisation but cannot rank close models.
- *Simulation limits.* The simulations are neutral and isothermal with one forcing magnitude; buildings are blocks without vegetation; each neighbourhood repeats periodically, without upstream terrain; and only one horizontal slice is modelled. Pedestrian height lies four vertical grid levels above the ground, where surface friction is represented by wall functions with $z_0 = 0.01\,\mathrm{m}$. The targets are time averages over 3 to 12 h (0.9 h for one flagged case) and carry sampling error, and the LES is not validated against measurements in this study.
- Cases are modelled on their native grids. A 0.5 m grid has four times the cell density of a 1 m grid, so 3,000 sampled cells cover a smaller area of an idealized case.
- The pixel model is not tuned beyond the capacity check; it is a diagnostic of the representation, not a competitor to convolutional models.

X. Conclusion and Future Work

Three data corrections, for raster orientation, the open-ground mask and the per-case friction velocity, precede all modelling. On held-out city and layout groups, LightGBM predicts the open-ground mean wind with $R^2 = 0.876$ (realistic 0.787) and, like the random forest, significantly exceeds the nine-feature baseline; most of this gain comes from the physically motivated features, and tuning can raise or lower accuracy on unseen cities. On the frozen test set every nonlinear model exceeds the baseline for the mean wind, but the 19 realistic cases are too few to rank them. At field level, a pixel-wise model recovers the level of each field but only part of its pattern; a random split changes its score on unseen neighbourhoods little and more model capacity does not improve it, which points to the local feature representation as the main limit.

Phase 2 therefore moves to models that read the geometry directly. It will train a U-Net [37] with circular padding and a wind-aligned frame and evaluate it on the same grouped folds. It will also build a hybrid that takes the level from the case-level model and learns the pattern, or that conditions the network on the physics features through feature-wise linear modulation (FiLM) [38], and deep ensembles will be evaluated for uncertainty calibration. At case level, an ensemble of the leading models and a tuning objective that does not neglect rare, very open neighbourhoods are the first improvements to test, on the same grouped folds. The question Phase 2 must answer is whether the advantage of convolutional models reported under random splits holds on unseen city groups.

Code and Data Availability

Code, data tables and all outputs are available at https://github.com/mahir-m01/Urban-Microclimate-Modelling. The environment is pinned with `uv` (`pyproject.toml`, `uv.lock`), and 118 automated tests cover the loader, features, metrics and splits, including the uniform-height limit of the Kanda parameterisation. The split file is checked against a stored hash on every load, the target table never contains test cases, the evaluator rejects predictions for test cases outside the final scoring, and the test-scoring script refuses a second run unless the reason is recorded. Each stage folder contains the script that regenerates its tables and figures, and the reported scores are read from those tables. The UrbanTALES simulations are distributed by their authors under CC BY-NC 4.0 [3] and are not redistributed.

Author Contributions

Mahir Abdullah: design and coordination of the end-to-end pipeline; the audit of all 538 cases, the shared loader and the evaluation tooling; the case-level model comparison, the pixel-level model and the frozen test evaluation; writing of this paper.

Neeraj Singh: exploratory analysis of the vertical profiles, building rasters, pedestrian wind fields, forcing parameters and time series, checking data quality, wind direction and simulation stability; the combined master case table and model-ready case list.

Om Mishra: preprocessing of the vertical profiles; the grouped frozen splits, the evaluation metrics and the evaluator.

Aparna Singh: the physics-informed feature-engineering library and its tests; the first nested tuning of the case-level models; the feature-selection and PCA study comparing accuracy against the number of features.

Ayush Aryan: the first case-level model ladder, from linear models to gradient-boosted trees, and the feature-set ablation.

Acknowledgment

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